Muhammad Bin Tariq

tariqmbin@gmail.com | +92 311 4519280 | MuhammadBinTariq | Muhammad Bin Tariq

Education

Lahore University of Management Sciences (LUMS) *Sept 2023 – June 2027*

Bachelor of Science in Computer Science, Minor in Mathematics; ***cGPA: 3.98/4.00***

- **Current Standing:** ***Top 1%*** of the BS Class of 2027
- **Selected Coursework:** Advanced Topics in Machine Learning, Dynamic Programming and Reinforcement Learning, Deep Vision-Language Models, Deep Learning for Space Time and Graphs, Applied Probability, Real Analysis, Algebra

Experience

Research Assistant - Representation Geometry of LLMs *Lahore, Pakistan*

Dr. Muhammad Tahir; Center for Urban Informatics, Technology and Policy at LUMS *Jan 2026 – Present*

- Formulated a study to investigate whether correctness-associated geometric trends reported in recurrent latent reasoning models such as Huginn-3.5b extend to hidden representaions of general purpose LLMs.
- Demonstrated exploratory analyses with OLMo 7B and Qwen 4B instruct models, through development of a reproducible pipeline to extract raw layer representations on reasoning tasks (AIME, MATH, GSM8K).
- Extended the pipeline to extract token-aligned hidden representations for Gemma 3-1B-IT on GSM8K.
- Characterized representation geometry across model depth and reasoning phases using spectral metrics, identifying differences between correct and incorrect solutions.
- Trained a Transformer-based correctness probe on complete reasoning hidden representations, achieving 0.816 ROC-AUC on held-out prompts and retaining 0.770 ROC-AUC when evaluated on early-reasoning representations.
- Established a framework to compare correct and incorrect reasoning subspaces using SVD and principal angles across 26 layers, evaluating their orientation differences against 1,000 size-matched random partitions.

Research Assistant - Methylation and Expression of Arabidopsis lncRNAs *Lahore, Pakistan*

Dr. Zaigham Shahzad and Dr. Laraib Iqbal Malik; Department of Life Sciences at LUMS *March 2026 – Present*

- Developed a Python/Bash pipeline using Pandas and GNU Parallel to profile CG, CHG, and CHH methylation across 11,071 Arabidopsis lncRNA loci and 927 methylome datasets.
- Integrated genomic annotations using BEDTools; joint analyses identified 23.5% TE overlap and 19.6x higher odds of TE-like methylation in overlapping versus TE-free loci.
- Guided and reviewed an RNA-seq workflow using BED12-to-GTF conversion, gffread, and Salmon to quantify lincRNA expression.
- Helped correct indexing and aggregation issues, improving agreement with published mean gene expression from Spearman $\rho = 0.898$ to $\rho = 0.960$.
- Collaboratively analyzed 2,115 lincRNAs, finding negligible association between mean expression and TE-like methylation conservation (Spearman $\rho = 0.035$).

Pedagogical Student Partner *Lahore, Pakistan*

Dr. Imran Anwar; LUMS Learning Institute *Jan 2025 – May 2025*

- Co-authored an instructional mathematics manuscript with Yousuf Bashir, adapting selected problems from Matoušek's *Thirty-three Miniatures* into guided exercises and visual explanations linking linear algebra to algorithms, combinatorics, and geometry.
- Collaborated weekly with Dr. Imran Anwar to develop teaching materials that directly informed new projects and readings in a redesigned Linear Algebra Honors course; formulated plans for undergraduate mathematics outreach seminars within LUMS.

Projects

STIM4RM: Memorization-Aware Reward Shaping for LLM Reasoning

- Led development of a GRPO-based prototype for OLMo-7B-Instruct, combining VersaPRM reasoning-step scores with STIM token-level memorization penalties to discourage memorized shortcuts in mathematical reasoning.
- Built a custom Hugging Face TRL trainer and a pipeline for four rollouts per prompt; added concurrent, cached Infini-gram queries to connect model predictions with pretraining-corpus frequencies for reward shaping.

Grokking and Reverse Engineering

- Reproduced delayed generalization on modular addition and subtraction using a one-layer Transformer trained on only 30% of 12,769 input pairs; addition test accuracy rose from 1.6% at epoch 500 to 100% by epoch 1,500, while subtraction reached 100% by epoch 3,500.
- Traced the shift from memorization to a frequency-based solution, identifying four key components whose removal reduced test accuracy from 100% to 12.7%; found that the learned representation became increasingly sparse while the loss landscape grew approximately 670× flatter.

LLM Alignment with PPO, DPO, GRPO, and Verifiable Rewards

- Built custom LoRA alignment pipelines for SmolLM2-360M/1.7B covering reward modeling, SFT, DPO, PPO, GRPO, and GSM8K RLVR; adapted 1.57M of 1.71B parameters (0.092%) on 42,537 HH-RLHF pairs.
- Achieved 71.37% reward-model accuracy and reduced SFT perplexity from 7.14 to 5.62; GRPO led the 360M benchmark with a 56% win rate and lower KL than PPO (0.0066 vs. 0.0090), while RLVR revealed 87.5 - 100% reward-degenerate groups.

Continuous and Discrete Vision-Language Modeling

- Designed two SmolLM2-based multimodal architectures: a continuous connector mapping 49 frozen CLIP patch tokens into the LM for CIFAR-10 VQA (synthetic captions), and a discrete VQ-VAE mapping synthetic images to a unified vocabulary of visual tokens for both VQA and autoregressive image generation.
- Evaluated multimodal systems on image captioning, image generation, forgetting, and spatial coherence, demonstrating that the discrete VQ-VAE approach outperformed the continuous connector in both accuracy and efficiency.

Curriculum Learning for Autonomous Driving

- Implemented and ran HighwayEnv benchmarks comparing PPO, DQN, and a custom DQN across a four-stage driving curriculum; contributed to PPO formulation and developed simulations for HighwayEnv and MetaDrive.
- The team's curriculum PPO run recorded 85% mean success across four HighwayEnv test configurations with one random seed; the study identified reward exploitation and forgetting of earlier driving skills during curriculum transitions.

Honors and Awards

LUMS, **Dean's Honor List** *2023 – Present*

LUMS, **Full Merit Scholarship** *2023 – 2027*

LUMS, **Ranked 2nd university-wide by cGPA** *2024*

Teaching

Teaching Assistant | AI501 - Mathematics for AI *Aug 2025 – Dec 2025*

- Supported a graduate-level course with 100+ students; held weekly office hours and volunteered additional time for student support and supplementary tutorials outside scheduled hours.

Technical Skills

Programming Languages: C/C++, Python, Haskell, JavaScript/TypeScript, SQL

ML/AI Tools and Frameworks: PyTorch, Torchvision, HuggingFace, Transformers, trl, OpenAI Gym, Stable Baselines3